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DATA320 Writing Assignment 2

1. This paper follows a standardized structuring framework that many clinical research articles draw from for accurate representations of the populations they represent. In being peer-reviewed and studied at multiple institutions, placebo-controlled trials have been established to be the “gold standard” for experimentation of quantifiable effects. The background alerts the reader to placebo effects as they pertain to pain management research throughout the body. Methods surrounding the randomized placebo-controlled trial are important because of their objective analysis. The design of these trials, alongside participant/patient selection, interventions, and resulting examinations, was made to be an integral incorporation of this paper’s scientific arguments. The results section in particular showcases data that compares groups and reports on statistically significant effects of treatment. From here, discussions are utilized to interpret the findings within the context of what is assumed to be already known due to prior research and limitations from potential clinical trials (supporting figures and tables). Conclusive findings also seek to reiterate the idea that the placebo effect technique they’ve adopted (IPRCT) is useful.
2. This experimental design utilized hypothesis testing for measurable placebo effects in order to verify quantifiable differences within the clinical trials they set up. Patients who

experienced pain in areas like the heel or dental domain were randomly assigned to different groups for effective testing. The treatment group would be receiving the placebo treatment as an active intervention, while the control group remained in a stable and controlled condition with no treatment or care. The random nature of it meant that comparing outcomes across groups was reasonable and grounded in scientific plausibility. A design process such as this means that researchers can make progress. After comparing, one could analyze the results to understand whether the differences between the two were due to the placebo rather than natural recovery.

3. The key findings from this study maintained that the placebo treatments produced a statistically significant improvement in pain outcomes compared to control conditions, revealing that the placebo effect has a measurable and clinically meaningful impact on people's lives. This contributes to the idea and societal understanding that placebo responses are not merely imagined or trivial in nature, but rather important psychologically and physiologically. The study also strengthens arguments that clinical trials must carefully account for placebo effects when evaluating the validity of treatment plans in the future. Even though these results aren't simply generalizable to all clinical trials, they are still mostly applicable to pain-related interventions and outcome measures that deal with the things analyzed in this study. The only differences would lie in magnitude or importance levels across medical conditions and specific patient populations.
4. A major question that comes to mind for the authors of this study lies in how well the blinding procedures operated in their experimentation when it came to participants truly believing that they received the active treatment intended for them. In this situation in

particular, expectation plays a critical role in placebo effect believability, so differences in patient understanding undoubtedly influence outcomes. Another concern is whether the sample size was sufficient for the scope of this experimental design, as there is a large number of discrepancies that can arise between the details for each group. It's important for researchers to control for confounding factors (pain level differences) and apply these controls to any statistical analysis they proceed to do after the fact (accounted for).

Criticism would likely lie in critiques about whether the patient felt comfortable interacting with the conductors of this study as they received the treatment and control outcomes.

5. If I were tasked with designing an experiment to build on the current understanding of the placebo effect, I would build on studies such as this to design a multi randomized trials with active treatment groups and placebo groups beyond the ones listed. Double blinding would allow for the outcome effects to be maximized, and a no-treatment control group would help with hypothesis testing about expectations and transparency influences on placebo effect sizes. The sample would need to be large and diverse enough to be representative of the entire population, and the conditions analyzed would deal with high-intensity pain areas for patients beyond what was listed. Measuring many different types of outcomes and psychological troubles would help in examining underlying mechanisms within a reasonable scope
6. This research paper required me to pay close attention to detail and technical jargon surrounding statistics. It was dense and hard at times to unpack the methodologies, as I don't often read clinical trial literature. This rough experience was compounded by the fact that research papers stray away from comfort and comprehension standards as we

know them. To approach the reading and unpacking of this paper effectively, I focused on the abstract and summary discussions prior to getting into the full descriptions so that I could understand the big picture. I then uncovered methods to clarify the structure of the experiment alongside the resulting effect sizes (significance tests could be overwhelming).

7. If I were to present the main findings to a wide college-educated audience, I would likely present the findings in a shorter and more condensed way that utilizes the tables and figures within the article. The formatting of the presentation would be similar to the science presentations I have done in the past, and experimentation would follow suit so that practical implications and plain language could be incorporated. The placebo, treatment, and control groups would be clearly outlined with slides that were effective enough for visual understanding and translating of statistical ideas (intuitive and relatable).